School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Report

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Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Probabilistic Problem Being Addressed

The main probabilistic problem in our Device-to-Device (D2D) Edge Computing project is the uncertainty in task processing time (TPT). This means the exact time required to execute a task is not known in advance. Many existing systems assume a fixed or deterministic execution time, but this does not reflect real-world behavior. In practice, task execution time varies depending on factors such as input data size and task complexity.

If we only use an average execution time, heavy or complex tasks may take much longer than expected. This can block system resources, increase waiting time for other tasks, and reduce overall system performance. To address this, we treat task processing time as a random variable, denoted as

$$T \sim f_T(t)$$

where $f_T(t)$ represents the probability distribution of task execution time.

Understanding the Problem (Highway Analogy)

To visualize this, consider a highway toll booth. The booth may assume that every vehicle needs 10 seconds to pass. However, a large truck may take 60 seconds. This creates a traffic jam for all vehicles behind it. In the same way, in our network, a heavy task may be treated like a normal task. When this happens, it blocks the processing queue and delays other tasks. This example shows why a probabilistic approach is needed. The system must handle varying task execution times instead of depending on fixed and predictable values.

System Objective

The objective of our system is to maximize the Task Completion Rate (TCR), which is the fraction of tasks completed before their deadlines. This can be written as:

$$\text{TCR} = \mathbb{P}(C_i \leq D_i)$$

where C_i is the completion time of task i and D_i is its deadline.

To achieve this, we use a game-based probabilistic offloading approach. Each device estimates the probability that a task will finish before its deadline if executed locally or offloaded to another device. Based on these probabilities, devices choose an offloading strategy. Using game theory concepts such as Nash Equilibrium, devices make balanced decisions so that no single device becomes overloaded.

Primary Sources of Uncertainty

- **Task Processing Time (TPT):** The time it takes to complete a task can vary depending on the data and the task itself. For example, processing an image with lots of features or objects takes longer than a simpler image.
- **Queueing and Contention Delay:** When multiple tasks compete for the same limited resources, unpredictable waiting times can occur.
- **Device Heterogeneity:** Different devices have different computing capabilities and workloads. This causes variation in task execution times.
- **Transmission Delay:** Sending tasks to other devices or nodes introduces communication delays, which can fluctuate depending on network conditions.

To manage these uncertainties, especially the variability in task processing time, our system uses a probabilistic model (specifically, an exponential distribution). This assumption allows for more realistic scheduling and supports smarter task offloading decisions under uncertainty.

At this early stage of the project, our understanding of task execution uncertainty is still developing. For now, we mainly focus on variation in task processing time as the main source of randomness, and we expect to improve this understanding as we study the system more in later milestones.

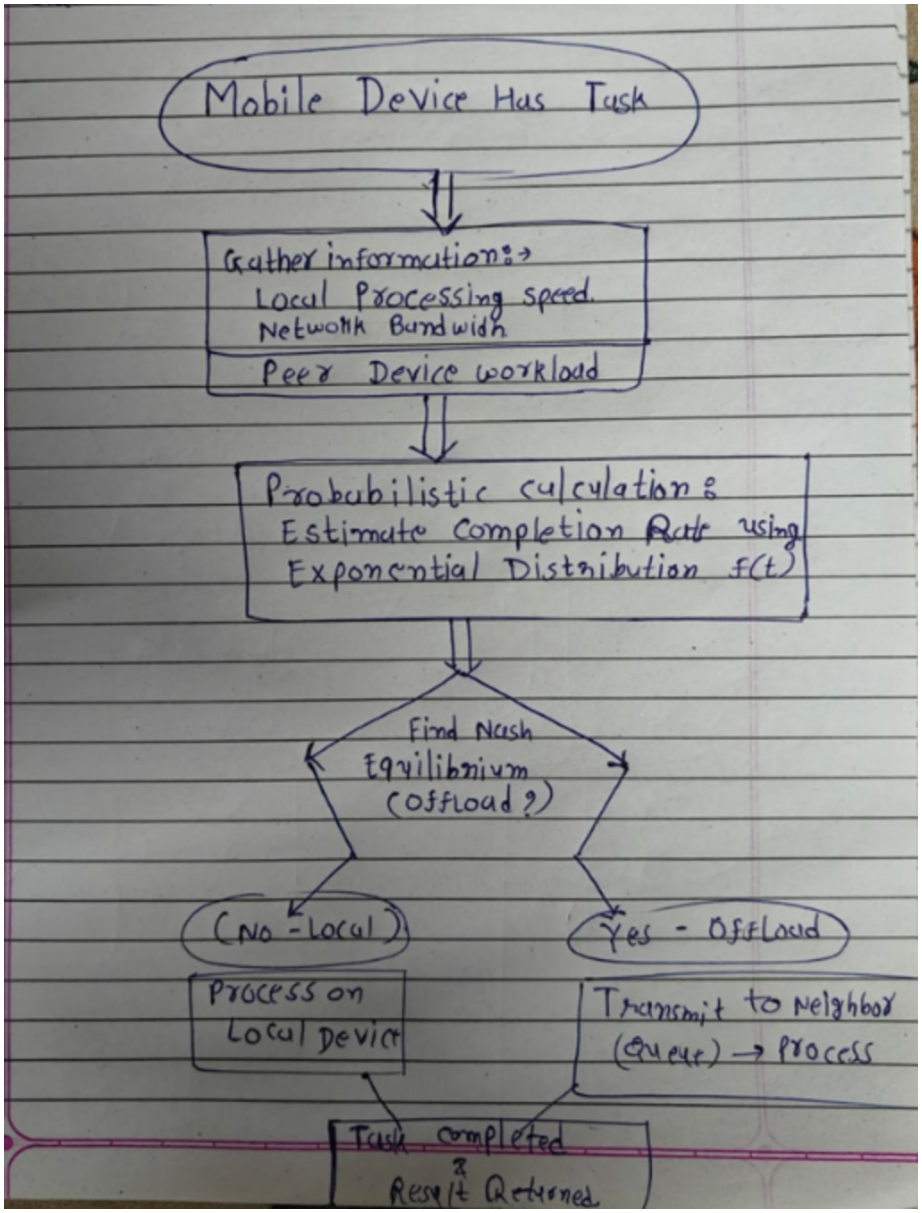

Figure 1: Flowchart for Q1: Task Offloading Decision Process

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

In our project, several aspects of the system are treated as random variables because their exact values cannot be known in advance.

Task Processing Time (TPT)

Task processing time is modeled as a continuous random variable. It depends on the application type and the size of the input data. Different applications behave differently, so we do not assume a single fixed execution time. Instead, task processing time is modeled as:

$$T_i \sim f_i(t)$$

where $f_i(t)$ is the probability distribution associated with application type i.

Task Arrival Process

The arrival of tasks is modeled as a Poisson process with arrival rate λ_k. This means tasks arrive randomly over time, and the exact arrival moments cannot be predicted. The probability of n tasks arriving in time t is:

$$P(N(t) = n) = \frac{(\lambda_k t)^n e^{-\lambda_k t}}{n!}$$

Task Completion Time

Task completion time is a random variable that includes:

- Queue waiting time (W),
- Task processing time (T),
- Transmission delay (D) if the task is offloaded.

It can be written as:

$$C = W + T + D$$

Since all these components are uncertain, the total completion time is also uncertain.

Task Completion Indicator

This is modeled as a Bernoulli random variable. It takes the value 1 if a task is completed before its deadline and 0 otherwise. This variable directly represents system performance.

Device Workload (λ_j)

Device workload represents the rate at which tasks arrive at a device. Since task arrivals change over time, the workload is dynamic and uncertain.

Uncertainty Modeling

Task Processing Time is represented using probabilistic distributions such as exponential, Gaussian, Gamma, or generalized extreme value distributions. The choice of distribution depends on the behavior of different application types.

- **Offline profiling:** Used to identify a suitable distribution for each application based on historical execution data.
- **Online parameter calibration:** Updates the distribution parameters using a small number of runtime samples. This allows the model to adapt to changes while keeping the computation overhead low.

At this stage, these distributions are chosen to make analysis easier and have not yet been fully validated using real execution data.

Current Probabilistic Assumptions

To simplify analysis and decision-making at the current stage of the project, we make the following assumptions:

- Task arrivals are independent, meaning the arrival of one task does not affect others.
- Service times for tasks of the same type are independent and identically distributed (i.i.d.).
- System conditions remain relatively stable over short time periods, allowing reliable probability estimates.

These assumptions are used to keep the initial model simple and easy to compute. As we collect more data and gain better understanding, these assumptions may be changed in later stages of the project.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

In our project, probabilistic reasoning plays an important role in how devices make task offloading decisions. The system carefully considers dependence, conditional probability, and independence to model interactions between devices and support efficient decision-making.

Dependence Between Devices

Devices in a Device-to-Device (D2D) network are dependent on each other because they share limited computational resources. When one device offloads a task to another device, it increases the workload and queue length of the target device. This affects the probability that future tasks assigned to the same device will be completed before their deadlines.

For example, if several devices offload heavy tasks to the same device, the task completion rate of all those devices becomes linked. The performance of one device is influenced not only by its own decisions but also by the decisions made by others. To handle this interaction, the system uses game-theoretic reasoning, where each device considers the congestion effects created by other devices. As a result, overall system performance emerges from collective behavior rather than isolated decisions.

Conditional Probability for Decision-Making

The offloading algorithm relies on conditional probability to make informed decisions. Instead of only considering whether a device is generally fast, the system estimates the probability that a task will be completed before its deadline given a specific offloading target:

$$P(\text{Success} \mid \text{Target} = j)$$

This probability depends on factors such as the current queue length, processing speed, and workload of the chosen target device. By conditioning on the selected target, the system can avoid congested devices and choose the most suitable one based on current network conditions.

For example, the processing time of a task depends on its input data. A photo containing one face can be processed quickly, while a photo containing many faces takes longer. Using probabilistic modeling (such as an exponential distribution), the system estimates the likelihood that the task will finish on time.

Independence Assumptions to Simplify Analysis

To keep the system efficient and suitable for real-time decision-making, some independence assumptions are made:

- **Independent task arrivals:** The arrival of one task does not affect the arrival of other tasks.
- **Memoryless service times:** Since task processing time is modeled using an exponential distribution, the remaining processing time does not depend on how long the task has already been running.

These assumptions simplify queueing analysis and allow efficient computation of task completion probabilities. An M/G/1 queueing model is used to relate task processing time distributions to overall task completion time, with probabilities derived using cumulative distribution functions rather than fixed time limits.

Decision-Making and Nash Equilibrium

Because devices influence each other through shared resources, offloading decisions are updated iteratively. The process continues until a Nash Equilibrium is reached. At this point, each device has selected the best possible strategy given the choices of other devices, and no device can improve its task completion probability by changing its decision alone.

At this early stage of the project, probabilistic reasoning and independence assumptions are used to keep the decision-making process simple and practical. Some system details are currently simplified, and we expect to improve these models as we better understand device interactions and workloads in later milestones.

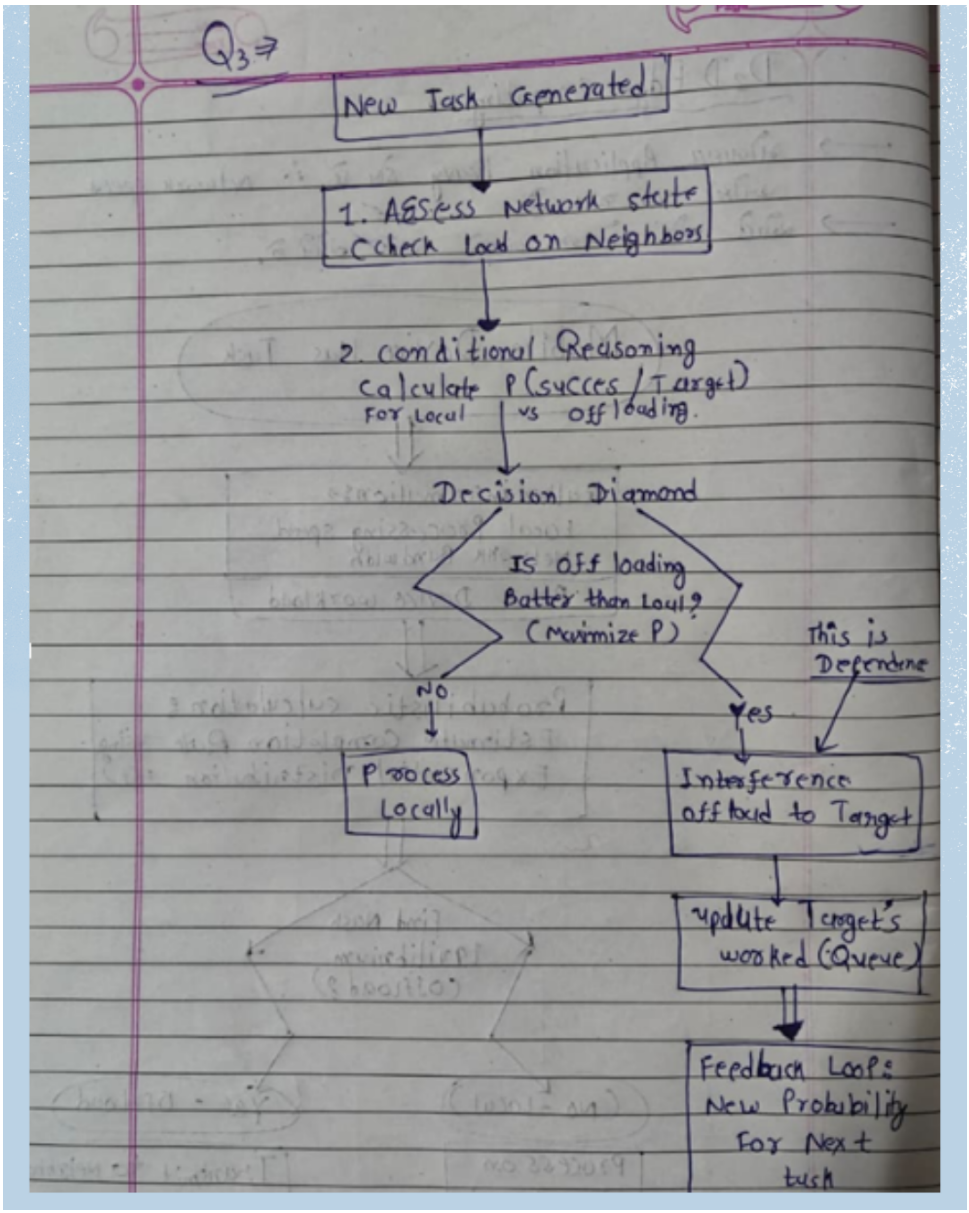

Figure 2: Flowchart for Q3: Task Offloading Decision Process

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

At this stage, our group has mainly read the paper to understand how the probabilistic model works. The key idea is that task processing time is uncertain. Because this time is not fixed, probability distributions are used instead of fixed average values.

Task Processing Time (TPT) Model

Different types of tasks can take different amounts of time to complete. Therefore, each application type is associated with a probability distribution. For example, image-processing tasks may use an exponential distribution, while speech-processing tasks may use Gamma or Gaussian distributions. These distributions describe how task execution time can vary.

Completion Rate Metric

The task completion rate is the probability that a task finishes before its deadline. This probability is calculated using task processing time distributions along with M/G/1 queueing theory. M/G/1 is a model where tasks arrive randomly, execution times vary, and each device processes one task at a time. The completion rate is used both for making offloading decisions and for evaluating performance in experiments.

Queueing Assumptions

- Tasks arrive randomly over time, which is modeled using a Poisson process.
- Each device has a single CPU and can execute only one task at a time. This follows the M/G/1 queueing model.

Direct Use in the Algorithm

In the paper, the MATURE offloading algorithm uses the probability that a task will finish before its deadline to decide where the task should be executed. This shows how the probabilistic model is directly applied in the algorithm.

Key Assumptions Affecting Design and Evaluation

- **Poisson Task Arrivals:** Tasks are assumed to arrive randomly, which allows the use of queueing theory and matches the experimental setup.
- **Single-Server Processing:** Each device is treated as a single server that handles one task at a time. This simplifies both analysis and implementation.
- **Known TPT Distributions:** Each application type is assumed to have a known execution time distribution, based on past data.
- **Completion Rate Instead of Average Delay:** The system focuses on completing tasks before deadlines rather than only reducing average execution time.
- **Independent but Interacting Devices:** Each device makes decisions independently, but these decisions can affect other devices by changing their workloads.

Since this is the first milestone, our understanding is still developing. At this stage, we know that task processing time uncertainty is important, completion rate is the main performance measure, and queueing theory links task behavior to overall system performance.

Example

If a task usually takes 5 seconds, a fixed-time system may reject it. A probabilistic model considers the chance that the task could finish faster, such as in 2 seconds. The algorithm uses this probability to make better offloading decisions.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

At this stage, our group has mainly read and understood the paper. The probabilistic model is used to handle uncertainty in Task Processing Time (TPT). Instead of assuming a fixed execution time for each task, the model represents TPT using probability distributions.

Let

$$T \sim f_T(t)$$

where T is the task processing time and $f_T(t)$ is its probability distribution.

Well-Understood Components and Relationships

Task Processing Time (TPT) Model:
Each application type is associated with a specific probability distribution.

$$T \sim \text{Exponential}(\mu) \quad \text{(Image-processing tasks)}$$

$$T \sim \text{Gamma}(k, \theta) \quad \text{or} \quad T \sim \mathcal{N}(\mu, \sigma^2) \quad \text{(Speech-processing tasks)}$$

These distributions capture variability in execution time across different tasks.

Completion Rate Metric:
The completion rate is the probability that a task finishes before its deadline D:

$$P(\text{Success}) = P(T_{\text{total}} \leq D)$$

Here, T_{total} includes both queueing delay and processing time. This probability is computed using M/G/1 queueing theory, which models random task arrivals, general service-time distributions, and a single processing unit. This metric directly supports deadline-aware task offloading decisions.

Basic Assumptions

- **Poisson Task Arrivals:** Task arrivals follow a Poisson process with rate λ:

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

- **Single-Server Processing (M/G/1):** Each device processes one task at a time on a single CPU.
- **Deterministic Transmission Time:** Communication delay is assumed to be fixed and does not vary.

Algorithmic Use in the System

The offloading decision algorithm (as described in the paper) evaluates the completion probability for each possible execution target j:

$$P(\text{Success} \mid \text{Target} = j)$$

The task is offloaded to the device that maximizes this probability. Devices update their choices iteratively until no device can improve its completion probability, leading to a Nash Equilibrium.

Aspects Expected to Change in Future Milestones

Since this is an early stage, several modeling choices may be refined later.

- **Task Arrival Process:** The Poisson arrival assumption may be replaced with bursty or time-varying arrivals:

$$\lambda = \lambda(t)$$

- **TPT Distribution Selection:** Instead of fixed distributions, the system may learn task models dynamically:

$$f_T(t) \leftarrow \text{Learned from data}$$

- **Transmission Modeling:** Transmission time may become stochastic:

$$T_{\text{tx}} \sim g(t)$$

- **Parallelism and Multi-Core Devices:** The M/G/1 model may be extended to multi-server queues such as M/G/k.
- **Parameter Adaptation:** Service rate parameters like μ may adapt based on workload:

$$\mu = \mu(\text{load}, \text{device state})$$

Justification

At this stage, our understanding is still exploratory. We clearly understand the core structure of the model, how uncertainty in task execution time is handled using probability distributions, and how completion probability guides offloading decisions. The listed components are expected to evolve because real systems are more complex than the simplified assumptions used initially. Future milestones will refine these models using experiments, learning, and deeper analysis.

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

At this stage, our understanding of the probabilistic aspects of the project is still developing. Several open issues remain, and some modeling choices are not yet fully resolved.

1. Choice of Probability Distributions for Task Processing Time (TPT)

Different applications are modeled using different probability distributions. For example:

$$T \sim \text{Exp}(\mu) \quad \text{(Image tasks)}$$

$$T \sim \text{Gamma}(k, \theta) \quad \text{or} \quad T \sim \mathcal{N}(\mu, \sigma^2) \quad \text{(Speech or complex tasks)}$$

However, we currently do not have a systematic rule or algorithm to select the correct distribution for a new or unknown application. At this stage, distributions are chosen based on assumptions from the paper rather than automatic classification or learning.

2. Reliability of Learned Parameters (μ)

The service rate μ represents how fast a device can process tasks. It is estimated from observed execution times as:

$$\mu = \frac{1}{\mathbb{E}[T]}$$

When workload or device conditions change, this estimate may no longer be accurate. It is still unclear:

- How often μ should be updated
- Whether online learning or periodic re-estimation should be used

This makes parameter stability an open issue for future milestones.

3. Poisson Task Arrival Assumption

The model assumes that tasks arrive according to a Poisson process with rate λ:

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

This assumption simplifies queueing analysis, but real task arrivals may be bursty or correlated. If arrivals are not Poisson, then predicted queue delays and completion probabilities may become inaccurate.

4. Simplified Transmission Time Modeling

Currently, transmission delay is modeled as a fixed value:

$$D_{\text{tx}} = \text{constant}$$

In reality, transmission delay is random and depends on network conditions. A more realistic model would treat it as a random variable:

$$D_{\text{tx}} \sim f(d)$$

This simplification is acceptable for early analysis but may limit realism.

5. Information Sharing Between Devices

The model assumes that devices have perfect and instant knowledge of system state, such as queue length and workload:

$$\text{Device } i \rightarrow \text{knows } \lambda_j, \mu_j \ \forall j$$

In real systems, this information may be delayed or incomplete. How limited or delayed information affects offloading decisions is still an open question.

Responsibility Allocation for the Next Milestone

In our five-member team, responsibilities follow Milestone 1 roles, with collaboration across tasks.

- **Scribe team (2 members):** Documents open issues, assumptions, and probabilistic reasoning.
- **Video/PPT team (2 members):** Explains probabilistic models, equations, and assumptions clearly in the presentation.
- **Concept map / GitHub member (1 member):** Visualizes probabilistic relationships, queueing models, and system interactions, and manages files.

All team members contribute ideas, review work, and give feedback to maintain shared understanding.

Looking Ahead

The open issues identified in this milestone will be addressed in later milestones through:

- More accurate probability models
- Algorithmic parameter updates
- Experimental validation

Documenting these uncertainties early helps us improve our assumptions and make the probabilistic model stronger as the project moves forward.

End of Scribe Report Submission